

Ameya Mahesh Chandwadkar

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SKILLS

Languages: C, Python, JavaScript, SQL

Computer Vision: OpenCV, Image Processing, Camera Calibration(6DOF)

Machine Learning and AI: Pytorch, Tensorflow, Scikit-learn, RAG, Model Optimization (Quantization), API based Systems

Data & Databases: NumPy, Pandas, ChromaDB, Vector Embeddings

WORK EXPERIENCE

Athena Infotech, Intern

Jan 2026 - Present

Machine Vision Engineer

Working on a hybrid C++/Python AI-powered vision pipeline, achieving 100+ FPS by implementing custom 3D triangulation in C++ for real-time intelligent perception.

PROJECT WORK

- **Plant Disease Detection:** Created a diagnostic system using deep learning with Convolutional Neural Networks (CNN) to identify more than 48 crop diseases from leaf images. Built a preprocessing pipeline for image normalization and enhancement using OpenCV. Launched the model through a Flask web API to allow real-time image uploads and diagnostic reporting. Python, TensorFlow/Keras, Flask.
- **Object Detection System for Drone:** Created a pose estimation module based on computer vision for a drone used for disaster relief that needs to navigate with extreme precision. The cv2.solvePnP algorithm was used to estimate the real-time 6D pose (position and orientation) of markers, along with AprilTag detection and camera calibration. A live drone simulator interface was powered by streamed telemetry via a Flask backend, allowing precise GPS-independent navigation.
- **Knowledge-Grounded Retrieval-Augmented FAQ System:** Knowledge-grounded FAQ chatbot built using TF-IDF and Logistic Regression for intent classification, combined with cosine similarity-based retrieval for response selection. The system follows a lightweight RAG-style architecture, where relevant information is retrieved from a structured knowledge base before generating a formatted reply
- **OpenCV Motion Detector:** Created a real-time motion detection system with OpenCV's Background Subtraction method. To separate moving regions, the pipeline applies thresholding, dilation, frame differencing, and grayscale conversion. Bounding boxes are drawn around detected objects and significant motion is identified using contour detection with area filtering.
- AND MORE...

EDUCATION

Thadomal Shahani Engineering College, Mumbai

B.E. in Computer Engineering

June 2024 - present

CGPA: 9.08/10

Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Data Mining, Advance Data Structures and Algorithms, Information Retrieval, Image Processing